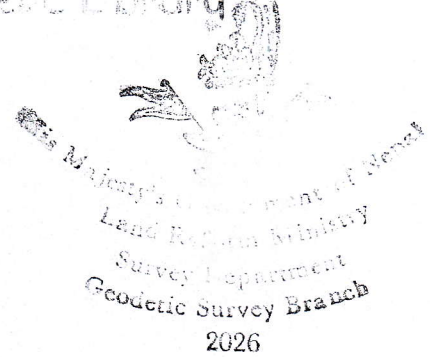


**REPORT AND RESULTS OF
GRAVITY SURVEYS IN NEPAL
1981-84
DIRECTORATE OF MILITARY SURVEY
MINISTRY OF DEFENCE U.K.**



REPORT AND RESULTS OF
GRAVITY SURVEYS IN NEPAL

1981-84

DIRECTORATE OF MILITARY SURVEY

MINISTRY OF DEFENCE UK

TWB
Geodesy Division
Mapping and Charting Establishment RE
December 1985

Reference: MCE/Caj

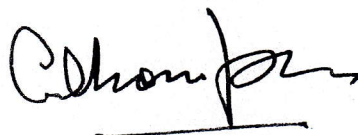
FOREWORD

This report, together with its companion report on the geodetic survey, gives in considerable detail the results of geodetic and gravity surveys carried out in the Kingdom of Nepal between 1981 and 1984 by military surveyors from 512 Specialist Team Royal Engineers and 19 Topographic Squadron Royal Engineers. The work was done in close cooperation with the Nepal Survey Department, and has provided a lasting high accuracy framework through the Kingdom on which any future survey and mapping work can be based.

The geodetic survey may well be the last national framework to be observed in the "classical" manner with theodolites and distance measuring equipment, for satellite based and other techniques will soon supersede earth based geometric figures in quality of results and obtainable at greatly reduced costs and time.

The mathematical adjustments presented in these reports are the work of the civilian staff of the Mapping and Charting Establishment Royal Engineers. The reports and analysis have been prepared by Mr T W Bassett, under whose expert guidance the work has been painstakingly carried out.

The high quality of the results of the surveys are a tribute to the demanding standards which were set and maintained both in the fieldwork and its subsequent adjustment, and the close cooperation between the surveyors of the United Kingdom and the Kingdom of Nepal.



C N THOMPSON
Major General
Director of Military Survey
United Kingdom

15 July 1986



SUMMARY

This report presents the results of gravity surveys made in Nepal
by British military survey units during the period 1981-84

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REPORT AND RESULTS OF GRAVITY SURVEYS IN NEPAL, 1981-84
DIRECTORATE OF MILITARY SURVEY, MINISTRY OF DEFENCE UK

INTRODUCTION

1. During the period 1981-84 British military survey units under the direction of the Directorate of Military Survey, Ministry of Defence UK (MODUK) were engaged in establishing a geodetic survey control net in Nepal to aid the Nepal Survey Department (NSD). As part of the aid gravity surveys were made to establish a gravity base station net and to observe gravity values for detail stations. This report presents the final results of the gravity surveys.

2. The periods of work in Nepal were:

<u>Season</u>	<u>Period</u>
1981	December 1980 to May 1981
1982	September 1982 to January 1983
1983	September 1983 to February 1984
1984	October 1984 to March 1985

In the 1981 season the unit concerned was No 512 Specialist Team Royal Engineers (512 STRE). Its primary task was to make Doppler satellite observations to position control stations for the geodetic survey net but a Fundamental Gravity Base Station and a small net of gravity base stations was established, and a few detail stations were observed in the vicinity of KATHMANDU. In the following three seasons the unit concerned was 19 Topographic Squadron (19 Sqn) of 42 Survey Engineer Regiment. The unit's primary task was to make ground survey observations of the geodetic control net but, particularly in the 1983 season, a significant effort was devoted to extending the gravity base station net and to observing detail stations wherever adequate mapping control for position and elevation was available.

3. Following the 1981 season of work a report entitled "RESULTS OF THE NEPAL GRAVITY SURVEY BY 512 STRE, 1981" was issued by Mapping and Charting Establishment RE in July 1982. Though there are no major changes the results given in the earlier report should be regarded as superseded by the data given in this present report.

GRAVITY REFERENCE SYSTEM

4. The reference system used for the gravity values presented here is IGSN71, Modified 1979 ie the International Gravity Standardization Net 1971 as modified in accordance with IUGG Resolution No 15, 1979 by removal of the Honkasalo correction from the published values of the original net. [Because the Honkasalo correction was retained in the IGSN71 reference system used for the July 1982 report the gravity values given there are smaller by about 0.03 mGal than the present values for stations.]

GRAVITY METER STANDARDIZATION

5. The instruments used for all gravity observations of the 1981-84 surveys were LaCoste Romberg Model G gravity meters. Two meters G142 and G191, used only in the 1981 season, were standardized to IGSN71 scale by observations over a 1670 mGal interval of the USA Mid Continent Calibration Line in November 1980. A further two meters used in the 1981 season were standardized, with reference to the accepted scale factors of the first two meters by equating the overall magnitudes of the total sets of corresponding gravity-differences observed with

each of the four meters. In the following seasons none of a group of four meters used were standardized directly over an independent gravity control interval but, in each successive season, a similar process of equating observed gravity-differences to fit corresponding values observed in previous seasons was used to achieve consistent scale.

6. The scale factors of the gravity meters derived for the various seasons are shown in the following table:

Season	G142	G191	G154	G383	G384	G435
1981	1.00037	1.00039	-	-	1.00054	1.00062
1982	-	-	1.00049	1.00073	1.00073	-
1983	-	-	1.00034	1.00055	1.00056	1.00038
1984	-	-	1.00064	1.00074	1.00060	1.00056

The standard-error (se) uncertainty of the scale factor determined for each of meters G142 and G191 was of the order of ± 0.00005 . The se uncertainties of the scale factors derived for the other four meters are estimated to be no worse than ± 0.00010 . The variations in scale factor values for each meter between the four seasons may reflect the noise level of the individual determinations rather than indicate an actual change of value. The overall range of gravity in the Nepal base station net is about 500 mGal so that the maximum effect of scale uncertainties on the mean observed gravity values of the base stations is not likely to be more than ± 0.02 or ± 0.03 mGal.

REDUCTION OF OBSERVATIONS

7. All gravity meter readings of observations in the survey have been reduced to milligal units using the relevant conversion table for the meter concerned, and the relevant scale factor quoted at paragraph 6 above. The observations were corrected for the effect of earth tides. *TIDE*

8. Gravity transfer trips to establish base stations were observed by the Ladder Sequence method so that each station of the trip was re-occupied for repeat readings to control gravity meter drift. At least two meters were used on each base net trip. Trips to fix detail stations from a reference base station were usually observed by a modified Ladder Sequence method so that at least two stations of the trip were re-occupied to obtain drift control. In each case the set of observations for a trip was solved, using the principle of least squares to accommodate redundant observations, on the basis that the gravity meter concerned had a constant linear rate of dynamic drift for the duration of the particular trip. The solution gave a value of the gravity-difference of each station from the initial reference station of the trip, together with an estimate of the se uncertainty of the value. *DRIFT*

NEPAL FUNDAMENTAL GRAVITY BASE

9. Prior to the 1981 season the Nepal Survey Department constructed a monument at Tribhuvan Airport, Kathmandu to serve as the Nepal Fundamental Gravity Base Station; this station is designated KATHMANDU J. On 23/24 May 1981 a gravity transfer was made from station KATHMANDU J to an IGSN71 station 06230A BANGKOK. The sequence of visits was KATHMANDU/BANGKOK/KATHMANDU and included observations at a temporary drift control station in Bangkok: the trip occupied a period of 33 hours including a 13 hour layover in Bangkok. Four gravity meters were used and the observed Δg -values obtained were:

(KATHMANDU J minus 06230A BANGKOK)

Meter	G142	G191	G384	G435	Mean
observed Δg	+361.106	+361.166	+361.131	+361.073	+361.12 mGal
se	$\pm .041$	$\pm .041$	$\pm .058$	$\pm .058$	$\pm .034$

Thus in the IGSN71, Modified 1979 system we have:

$$\begin{aligned} 06230A \text{ BANGKOK } g &= 978\,300.10 \pm 0.032 \text{ mGal} \\ &\quad +361.12 \pm 0.034 \\ \text{KATHMANDU J } g &= 978\,661.22 \pm 0.047 \end{aligned}$$

10. In the period July–October 1979, 512 STRE carried out a gravity survey in Sri Lanka. During the course of that survey, and in anticipation of the 1981 survey in Nepal, gravity transfers were made in October 1979 from Sri Lanka to a station KATHMANDU JA in Nepal using three gravity meters on two separate trips; the latter station was connected to station KATHMANDU J during the 1981 survey. By combining mean observed Δg -values from the two surveys we have in the IGSN71, Modified 1979 system:

$$\begin{aligned} 02969J \text{ COLOMBO } g &= 978\,116.94 \pm 0.053 \text{ mGal} \\ 1979 \Delta g &\quad + 546.06 \pm 0.05 \\ \text{KATHMANDU JA } g &= 978\,663.00 \pm 0.07 \\ 1981 \Delta g &\quad -1.97 \pm 0.004 \\ \text{KATHMANDU J } g &= 978\,661.03 \pm 0.07 \end{aligned}$$

The g -value for KATHMANDU J by transfer from COLOMBO differs by -0.19 mGal from that obtained by the transfer from BANGKOK. However it must be noted that none of the three meters (G383, G384 and G435) used in 1979 were standardized by direct measurement of a gravity control interval and, in the reduction of the observations, all the meters were assumed to have a scale factor of unity. Now scale factors for the same three meters during the 1981–84 survey were found to be of the order of 1.00060 and if it is assumed that similar values should have been applied in 1979 then the mean Δg value of $+546.06$ mGal from COLOMBO to KATHMANDU would increase by about $+0.33$ mGal. Thus the difference in g -value for KATHMANDU from the BANGKOK and COLOMBO transfers would become $+0.14$ mGal instead of -0.19 mGal.

11. It would be unwise to place much reliance on the g -value for KATHMANDU J derived from the 1979 transfer from COLOMBO but it does indicate that the 1981 transfer from BANGKOK contains no significant systematic error. The fundamental gravity value for Nepal is therefore accepted to be, from the 1981 transfer:

$$\text{KATHMANDU J } g = 978\,661.22 \pm 0.047 \text{ mGal}$$

GRAVITY BASE STATION NET

12. A diagram of the base net established in Nepal showing the various connections observed is given at Annex D; 7 excentre bases in Kathmandu fixed directly from the fundamental base are not shown in the diagram. Ties to other bases of the net were observed from one of the four interconnected bases KATHMANDU J, JA, D and DB.

13. The main net contains 45 stations at 35 different locations and these were connected by 268 observed Δg -values on 69 lines. To accommodate redundant observations a least squares adjustment of the observational data was made, holding fixed the accepted g -value of station KATHMANDU J. The *a priori* estimates of se uncertainty assigned to the various observations had values of

either ± 0.02 or ± 0.03 mGal depending on the reliability of an observation indicated by the internal consistency of the relevant transfer trip. The statistical analysis of the adjustment solution showed that the estimates of uncertainty were satisfactory.

14. A list of the adjusted g-values of the base stations is given at Annex A. The Annex also shows the number of ties from each station to other stations of the net. Two values of the se uncertainty of an adjusted g-value are given; first an uncertainty relative to the fundamental base station as origin to indicate the reliability of the base net observations and second, a total uncertainty in the IGSN71 reference system. Relative to the fundamental base station the maximum se uncertainty in the g-value of a base station is ± 0.037 mGal.

15. The gravity base stations are sited either at airports or at local centres of road communication; a description sheet for each base station is given at Annex F. The position and elevation value quoted for each base station at Annex A has been taken from the best available map source (see paragraph 19). A two-digit numerical code is appended to each position and elevation value to indicate its reliability and an explanation of these codes is given at Annex C.

GRAVITY DETAIL STATIONS

16. Values of observed gravity were determined at a total of 375 detail stations as listed at Annex B. The distribution of these stations is shown in the diagram at Annex E.

17. Most of the trips to fix detail stations from a reference base station were made using a single gravity meter so that no reliability can be quoted for the g-values. However, from the general standard of observations indicated by the drift rates of the meters during the various trips, and the evidence of agreement at those stations where repeat observations were made, the se uncertainty of the g-values is unlikely to exceed ± 0.3 mGal relative to the IGSN71 reference system.

18. The number and distribution of the detail stations established was limited by the availability of survey and mapping data to provide the necessary position and elevation values of reasonable accuracy - bearing in mind, particularly, the desirable accuracy level of no worse than ± 3 metres in elevation. Where possible detail stations were sited at benchmarks of the Nepal Level Net which, at present, is confined to lines through the plains of the Terai east of Bhairawa ($27^{\circ}30'N$, $83^{\circ}25'E$) and a line to Kathmandu. For some of the benchmarks occupied it was found that, as yet, NSD were not able to supply levelled elevation values.

19. For the majority of detail stations the position and elevation values quoted at Annex B have been taken from the best available map sources. The principal map series used were:

<u>Title</u>	<u>Scale</u>	<u>Contour Interval</u>
Kathmandu Valley	1:50,000	5 metre
NSD Land Systems	1:50,000	No contours
Survey of India (GSGS 5501)	1:63,360	100 feet (50 feet in plains)
NSD Geological	1:125,000	100 feet

Elevation values taken from the maps with a contour interval of 100 feet could only be estimated to within ± 50 feet; where this is the case corresponding

values quoted in metre units have been rounded to the nearest 10 metres. The reliability codes of position and elevation values annotated in the listing at Annex B are those explained at Annex C.

CONCLUSIONS

20. The 1981-84 gravity surveys in Nepal have established an adequate datum g-value for the Nepal Fundamental Gravity Base and a sound net of gravity base stations covering most of the more accessible parts of the country.

21. Though some detail survey has been done its usefulness in providing gravity anomaly values of an accuracy better than ± 5 mGal is limited by a general lack of reliable elevation data.

NEPAL GRAVITY SURVEYS 1981-84

GRAVITY BASE STATIONS

IGSN71 SYSTEM (Modified 1979)

Station		Position		Relia- bility Code	Elevation	Relia- bility Code	Observed g-value	Standard Local	Error IGSN	No. of Ties
		Lat N	Long E							
<u>Fundamental</u>					metres		mGal		±	
<u>Base</u>										
KATHMANDU	J	27 41.82	85 21.62	10	1332.1	10	978661.22	<u>datum</u>	<u>.047</u>	51
<u>Excentre</u>										
<u>Bases</u>										
KATHMANDU	C	27 42.4	85 19.2	22	1311	41	978664.50	±	±	6
"	CA	27 41.2	85 19.6	22	1290	41	667.35	.014	.049	4
"	CB	27 40.6	85 20.5	22	1304	41	662.87	.008	.048	4
"	CC	27 43.2	85 19.6	22	1335	44	661.72	.015	.049	4
"	D	27 42.4	85 19.0	22	1306	41	666.74	.037	.060	2
"	DA	27 42.4	85 19.0	22	1306	41	666.66	.014	.049	15
"	DB	27 40.7	85 18.7	22	1250	46	676.35	.020	.051	2
"	F	27 42.2	85 19.3	22	1297.6	21	666.54	.012	.049	119
"	JA	27 41.4	85 21.7	22	1337	41	663.19	.013	.049	4
"	K	27 41.5	85 19.0	22	1299	41	978667.82	.011	.048	23
								.008	.048	4
<u>Regional</u>										
<u>Bases</u>										
BAGLUN	J	28 15.7	83 36.6	23	950	46	978762.48	±	±	2
BARTUNG	E	27 51.5	83 33.4	24	610	46	776.21	.035	.059	2
BENI	D	28 20.1	83 34.1	23	820	46	772.72	.026	.054	2
BHADRAPUR	D	26 32.9	88 05.5	10	87	13	888.85	.035	.059	2
BHAIRAWA	D	27 30.2	83 27.0	23	110	46	901.09	.024	.053	4
"	J	27 30.3	83 25.0	24	110	46	903.19	.020	.051	4
BHARATPUR	E	27 42.0	84 25.7	23	180	46	879.90	.014	.049	46
BIRGANJ	C	27 00.9	84 52.7	22	82	11	873.05	.014	.049	16
BIRATNAGAR	J	26 29.1	87 16.2	24	80	46	878.35	.022	.052	3
BUTWAL	D	27 42.2	83 27.7	23	180	46	906.52	.016	.050	9
"	E	27 40.8	83 28.1	24	120	46	906.59	.015	.049	16
CHARIKOT	C	27 40.1	86 02.4	24	1900	46	521.60	.020	.051	5
CHAURJAHARI	J	28 37.0	82 26.5	25	760	46	871.95	.019	.051	5
"	JA	28 37.6	82 26.6	22	760	46	871.87	.013	.049	6
DHALKE	E	26 55.7	85 57.4	24	120	46	845.77	.026	.054	4
DHANGADHI	C	28 42	80 36	25	180	46	991.86	.025	.053	3
"	J	28 45.8	80 34.6	24	170	46	990.27	.033	.057	2
DHARAN	F	26 48.6	87 16.4	24	300	46	829.78	.017	.050	5
DOTI	J	29 15	80 57	25	700	46	929.45	.017	.050	25
HETAUDA	D	27 25.5	85 02.0	23	460	46	828.99	.022	.052	2
JANAKPUR	J	26 42.62	85 55.62	10	70.1	10	866.36	.017	.050	7
JUMLA	J	29 17.1	82 13.1	22	2320	46	543.53	.033	.057	4
KANCHANPUR	E	26 33.1	86 53.2	24	90	46	879.90	.021	.051	3
KODARI	E	27 58.3	85 58.0	24	-	-	497.16	.025	.053	3
LAMBI	E	27 53.0	82 30.4	24	300	46	942.92	.033	.057	2
LAMOSANGU	E	27 44.7	85 53.0	24	1140	46	751.63	.017	.050	11
LAUKI	E	27 59.0	82 10.6	24	300	46	960.53	.015	.049	11
MUGLIN	D	27 51.5	84 33.7	22	220	46	978870.74	.017	.050	9
								.013	.049	24

Station		Position				Relia- bility Code	Elevation	Relia- bility Code	Observed g-value	Standard Local	Error IGSN	No. of Ties
		Lat N	Long E									
							metres		mGal	±	±	
NABALPUR	E	27 04.0	85	35.4	24		150	46	978854.63	.025	.053	3
NAGARKOT	C	27 41.52	85	31.34	10		2164.2	10	490.52	.015	.049	19
NEPALGANJ	J	28 07	81	37	25		170	46	968.19	.015	.049	8
"	JA	28 06.1	81	40.1	22		155	43	965.42	.014	.049	25
POKHARA	F	28 13.9	83	59.4	23		890	46	771.29	.035	.059	2
"	J	28 12.1	83	59.0	25		820	46	784.90	.012	.049	7
"	JA	28 12.1	83	59.0	22		820	46	784.97	.017	.050	6
RAPTI RIVER	E	27 49.7	82	45.7	24		300	46	930.46	.019	.051	6
REUBENI	E	26 37.7	86	42.9	24		90	46	869.83	.029	.055	4
SIMRA	J	27 09.8	84	58.9	22		137	36	863.32	.015	.049	11
SURKHET	J	28 35.0	81	38.1	22		730	46	895.30	.024	.053	5
TAULIHAWA	E	27 38.8	83	03.1	24		150	46	919.78	.015	.049	12
TUMLINGTAR	J	27 19	87	12	25		520	46	978832.55	.035	.059	2

NEPAL GRAVITY SURVEYS 1981-84

GRAVITY DETAIL STATIONS

Detail Station	Position		Relia- bility Code	Elevation	Relia- bility Code	Observed g-value	No. of Ties
	Lat N	Long E		metres		mGal	
1981 from KATHMANDU J							
JK1- 1	27 41.3	85 21.0	22	1295.7	21	978666.82	2
JK1- 2	41.8	20.7	22	1322	41	978665.42	2
JK1- 3	42.4	20.9	22	1318	41	978663.61	4
JK1- 4	43.1	20.8	22	1334	41	978666.00	2
JK1- 5	43.7	20.3	22	1344.5	21	978660.05	2
JK1- 6	43.7	19.7	22	1313.6	21	978665.64	2
JK2- 1	27 40.5	85 21.1	22	1291.7	21	978666.72	4
JK2- 2	40.1	21.1	22	1309.1	21	978662.21	4
JK2- 3	39.6	20.5	22	1290	41	978666.63	2
JK2- 4	39.4	20.1	22	1296.5	21	978667.70	4
JK2- 5	38.8	19.7	22	1331	41	978663.26	2
JK2- 6	39.0	19.2	22	1327.4	21	978665.05	2
JK2- 7	39.3	18.6	22	1311.1	21	978672.24	2
JK2- 8	39.7	18.3	22	1300	44	978675.03	2
JK2- 9	40.4	18.0	22	1279.5	21	978684.94	2
JK2-10	41.0	17.0	22	1317.5	21	978676.37	2
JK2-11	41.8	17.0	22	1320	44	978678.89	2
JK2-12	42.5	17.3	22	1322.5	21	978674.27	2
JK2-13	42.8	18.0	22	1302	41	978670.12	2
JK2-14	43.0	18.4	22	1294	41	978671.33	2
JK2-15	43.5	19.0	22	1299	41	978670.49	2
JK3- 1	27 39.5	85 19.5	22	1316.4	21	978668.09	2
JK3- 2	39.7	19.0	22	1330.7	21	978673.97	2
JK3- 3	39.6	18.8	22	1319.8	21	978671.11	2
JK3- 4	40.1	18.7	22	1310.2	21	978676.16	2
JK3- 5	40.6	19.1	22	1283.8	21	978673.42	2
JK4- 1	27 40.6	85 20.2	22	1304	41	978663.93	2
JK4- 2	40.9	19.6	22	1288.8	21	978668.74	2
JK4- 3	41.7	19.5	22	1290.5	21	978667.65	2
JK4- 4	41.9	19.8	22	1305.1	21	978664.21	2
JK4- 5	42.4	19.9	22	1308.8	21	978664.30	2
JK4- 6	42.8	20.0	22	1306.3	21	978665.02	2
JK4- 7	43.3	19.8	22	1330.9	21	978660.86	2
JK4- 8	41.0	19.0	22	1285.3	21	978671.34	2
JK4- 9	41.1	18.0	22	1278.9	21	978676.16	2
JK4-10	40.8	18.1	22	1282.8	21	978679.22	2
JK5- 1	27 39.7	85 24.6	22	1310	44	978673.16	2
JK5- 2	40.1	26.6	22	1330	44	978659.42	2
JK5- 3	40.7	27.6	22	1340	44	978657.40	2
JK5- 4	41.6	28.4	22	1380	44	978659.12	2
JK6- 1	27 41.1	85 21.4	22	1317	41	978662.84	2
JK6- 2	40.3	21.1	22	1309.1	21	978663.40	2
JK6- 3	39.9	20.8	22	1304	41	978664.46	2
JK6- 4	39.0	19.9	22	1317.2	21	978664.88	2
JK6- 5	39.1	19.0	22	1328.6	21	978667.25	2

Detail Station	Position				Relia- bility Code	Elevation	Relia- bility Code	Observed g-value	No. of Ties
	Lat N	Long E				metres		mGal	
JK6- 6	27	39.9	85	18.2	22	1273	41	978677.39	2
JK6- 7		40.2		18.2	22	1280	41	978684.21	2
JK6- 8		40.5		17.7	22	1302	41	978682.25	2
JK6- 9		40.7		17.1	22	1295.1	21	978679.93	2
JK6-10		41.3		17.5	22	1294	41	978678.21	2
<u>1982 from KATHMANDU J</u>									
HTA1-1 BOPUNGSOKOPE	26	57.00	86	56.89	10	1782	13	978553.8	2
HTA1-2 DHAJE	26	51.95	87	23.69	10	2047	13	978492.2	1
HTA2-1 LENDRUNGCHULI	27	13.40	86	02.03	10	1898	13	978525.9	2
HTA3-1	26	53.2	87	20.0	23	1455	13	978617.8	2
HTA3-2	26	58	87	21	24	1104	13	978717.4	1
<u>1983 from KATHMANDU DB</u>									
HT1- 1 SIRANCHOK	28	05.02	84	36.18	10	1973	13	978509.3	2
HT1- 2 SIRAICHULI	27	45.73	84	38.00	10	1945	13	978518.5	2
HT1- 3 KUMARI	27	48.54	85	08.47	10	2068	13	978506.9	2
HT1- 4 BURICHAUR	27	30.6	85	09.8	10	2500	13	978400.4	2
HT1- 5	27	32.2	85	10.3	24	1910	46	978555.5	2
HT2- 1 BM 101-001	27	41.1	85	07.7	22	745.2	10	978775.9	2
HT2- 2 BM 101-006	27	49.9	85	03.4	22	568.6	10	978807.2	2
HT2- 3 BM 101-011	27	45.4	85	02.8	22	501.5	10	978823.0	2
HT2- 4 BM 101-021	27	48.1	84	59.6	22	336.1	10	978853.2	2
HT2- 5 BM 101-023	27	48.5	84	58.2	22	327.8	10	978855.8	2
HT2- 6 BM 101-033	27	48.3	84	53.2	22	329.7	10	978859.0	2
HT2- 7 BM 101-038	27	48.0	84	50.5	22	291.7	10	978865.7	4
HT2- 8 BM 101-050	27	48.1	84	45.1	22	242.6	10	978872.9	2
HT3- 1	27	43.2	85	08.6	22	820	46	978753.0	3
HT3- 2 BM 101-043	27	48.8	84	48.0	22	277.2	10	978862.7	2
HT3- 3 BM 101-047	27	48.4	84	46.5	22	267.7	10	978867.2	2
HT3- 4 BM 101-054	27	47.7	84	43.1	22	234.0	10	978870.8	2
HT3- 5 BM 101-062	27	50.4	84	39.8	22	210	46	978860.6	2
HT3- 6 BM 101-074	27	51.8	84	34.9	22	180.2	10	978866.5	2
HT3- 7 BM 101-066	27	51.4	84	38.6	22	200	46	978866.6	2
HT4- 1	27	28.6	84	44.4	23	350	46	978811.8	2
HT4- 2	27	46.5	84	13.7	23	1730	46	978586.6	2
HT4- 3 THAPREK	28	05.02	84	11.24	10	1281	13	978675.5	2
HT4- 4	28	08.9	83	44.3	23	1900	46	978575.9	2
HT5- 1	27	43.8	85	18.5	20	1296	32	978673.0	1
HT5- 2	27	47.4	85	16.8	20	1461	32	978645.5	1
HT5- 3	27	48.8	85	13.0	20	1619	32	978606.4	1
HT5- 4	27	48.3	85	16.2	20	1854	32	978566.4	1
HT5- 5	27	49.9	85	12.3	20	1870	32	978625.6	1
HT5- 6	27	51.2	85	12.6	20	1880	32	978676.5	1
HT6- 1	27	40.9	85	22.6	22	1295	44	978664.9	1
HT6- 2	27	40.5	85	25.6	22	1305	44	978673.6	1
HT6- 3	27	38.2	85	42.9	22	626	36	978797.2	1
HT6- 4	27	41.9	85	42.9	22	1220	46	978693.8	1
HT6- 5	27	44.9	85	44.8	22	1500	46	978604.5	1
HT6- 6	27	46.5	85	43.0	22	1418	36	978631.9	1
HT6- 7	27	39.1	85	36.7	22	871	36	978763.5	1

Detail Station	Position				Relia- bility Code	Elevation	Relia- bility Code	Observed g-value	No. of Ties
	Lat N	Long E							
	°	'	°	'		metres		mGal	
HT6- 8	27	37.0	85	33.4	22	1490	46	978629.7	1
HT6- 9	27	37.9	85	31.2	22	1463	34	978646.1	1
HT7- 1	27	30.4	85	03.3	23	910	46	978813.1	3
HT7- 2	27	32.8	85	08.0	23	1120	46	978716.1	1
HT8- 1	27	52.9	85	54.2	22	820	46	978716.4	2
HT8- 2	27	50.6	85	53.3	22	1000	46	978672.1	2
HT8- 3	27	47.3	85	55.0	22	1230	46	978630.9	2
HT9- 1	27	34.7	85	30.9	22	1447	34	978652.8	1
HT9- 2	27	34.9	85	22.8	22	1535	44	978636.3	1
HT9- 3	27	40.4	85	18.0	22	1280	44	978685.8	1
HT9- 4	27	38.2	85	16.7	22	1280	44	978682.5	1
HT9- 5	27	36.1	85	16.0	22	1570	34	978633.4	1
HT10-1 HUWAKOT	27	45.88	83	36.41	10	1843	13	978579.4	1
HT10-2 JOMSON AIRPORT	28	46.8	83	44.0	24	2680	46	978375.7	1
HT11-1	27	33.6	85	24.5	22	2762	34	978359.7	1
HT12-1 CHAITIDHURI	28	29.7	82	38.5	24	3050	13	978375.8	1
HT12-2 GHARTIGARA	28	40.67	82	15.71	10	1871	13	978621.5	1
HT12-3	28	45.3	82	07.6	24	1520	46	978701.2	1
HT12-4 BANSPANI	28	32.41	81	28.62	10	1585	13	978686.1	2
HT12-5 NIGALCHULA	28	20.37	82	03.28	10	2407	13	978519.8	1
HT12-6 RANJA	28	03.74	82	34.50	10	1706	13	978664.6	1
HT21-1	27	41	85	13	25	1150	est	978660.7	1
HT21-2	27	42	85	12	25	1100	est	978652.4	1
HT21-3	27	43	85	11	25	1300	est	978679.6	1
HT21-4	27	43	85	10	25	1450	est	978721.0	1
HT21-5	27	43	85	09	25	1050	est	978746.6	1
HT22-1 BM 114-009	27	44.1	84	27.3	23	270	46	978872.8	1
HT22-2 BM 114-013	27	46.0	84	28.0	23	300	46	978882.4	1
HT22-3 BM 114-018	27	47.3	84	26.0	23	210	46	978884.4	1
HT22-4 BM 114-022	27	49.0	84	27.0	23	180	46	978886.8	1
HT22-5 BM 114-026	27	49.1	84	29.0	23	240	46	978882.6	1
HT22-6 BM 114-030	27	49.6	84	31.4	23	150	46	978872.5	1
HT22-7 BM 114-034	27	50.2	84	33.4	23	180	46	978867.6	1
HT23-1 BM 100-190	27	43.6	85	06.3	23	1150	10	978695.1	1
HT23-2 BM 100-180	27	41.2	85	06.0	23	1568	10	978614.1	1
HT23-3 BM 100-179	27	40.9	85	04.9	23	1675	10	978593.1	1
HT23-4 BM 100-172	27	39.5	85	06.1	23	1863	10	978556.3	1
HT23-5 BM 100-168	27	38.7	85	05.7	23	1712	10	978583.5	1
HT23-6 BM 100-160	27	37.0	85	05.1	23	1939	10	978533.7	1
HT23-7 BM 100-155	27	36.7	85	05.8	23	2223	10	978471.0	1
HT23-8 BM 100-184	27	42.4	85	05.4	23	1461	10	978634.6	1
HT23-9 BM 100-194	27	43.5	85	07.5	23	969	10	978731.2	1
HT24-1 BM 100-132	27	35.1	85	05.0	23	2304	10	978452.7	1
HT24-2 BM 100-121	27	34.4	85	04.8	23	2031	10	978507.0	1
HT24-3 BM 100-117	27	33.5	85	04.2	23	1824	10	978551.7	1
HT24-4 BM 100-113	27	33.8	85	04.1	23	1647	10	978589.2	1
HT24-5 BM 100-105	27	32.9	85	02.0	23	1199	10	978683.7	1
HT24-6 BM 100-101	27	32.1	85	02.3	23	1104	10	978708.0	1
HT24-7 BM 100-096	27	32.0	85	02.8	23	825	10	978760.8	1
HT24-8 BM 100-091	27	31.0	85	03.2	23	622	10	978800.2	1
HT24-9 BM 100-083	27	28.4	85	02.9	23	431	10	978826.8	1
HT25-1	27	50.1	85	10.4	23	1190	46	978708.8	1

Detail Station	Position				Relia- bility Code	Elevation	Relia- bility Code	Observed g-value	No. of Ties
	Lat N	Long E							
						metres		mGal	
HT25-2	27	50.7	85	09.0	23	1280	46	978755.8	1
HT25-3	27	51.6	85	08.6	23	910	46	978808.5	1
HT25-4	27	53.1	85	08.8	23	490	46	978810.5	1
HT25-5	27	55.4	85	09.2	23	540	46	978811.6	1
1983 from BHAIRAWA J									
HT13-1	27	56.2	83	25.9	24	430	46	978842.1	2
HT13-2	27	52.4	83	28.1	24	1520	46	978702.4	1
HT13-3	27	48.2	83	31.1	24	550	46	978833.9	1
HT13-4	27	45.0	83	30.4	24	270	46	978888.4	1
HT14-1	27	32.5	83	03.1	24	100	46	978920.9	1
HT14-2	27	33.1	82	51.0	24	100	46	978919.6	1
HT14-3	27	38.8	82	52.0	24	130	46	978920.1	1
HT14-4	27	44.4	82	50.5	24	150	46	978924.2	1
HT14-5	27	41.5	83	10.2	24	130	46	978914.2	1
HT14-6	27	41.5	83	20.0	24	120	46	978913.5	1
HT15-1	27	36.3	83	38.6	24	130	46	978892.2	1
HT15-2	27	31.9	83	40.0	24	110	46	978889.2	1
HT15-3	27	36.8	83	57.3	24	170	46	978892.2	1
HT15-4	27	35.1	83	52.7	24	180	46	978898.6	1
HT15-5	27	33.8	83	43.0	24	120	46	978889.7	1
HT15-6	27	39.3	83	34.4	24	170	46	978898.8	1
HT15-7	27	27.0	83	17.0	24	90	46	978913.0	1
HT15-8	27	29.8	83	20.2	24	90	46	978908.3	1
HT15-9	27	33.2	83	47.7	24	120	46	978892.0	1
1983 from NEPALGANJ JA									
HT16-1	28	26.3	81	43.9	24	670	46	978908.5	1
HT16-2	28	21.4	81	42.9	24	300	46	978954.6	1
HT16-3	28	16.5	81	39.6	24	180	46	978964.7	1
HT18-1	28	11.9	81	41.5	24	160	43	978961.6	2
HT18-2	28	07.2	81	43.3	24	160	46	978968.7	1
HT18-3	28	08.2	81	53.1	24	210	46	978966.1	1
HT18-4	28	03.5	81	55.7	24	170	46	978968.1	1
HT18-5	28	09.6	81	45.2	24	150	46	978964.5	1
HT35-2	28	03.9	81	37.3	23	150	43	978966.8	1
HT35-3	28	01.1	81	36.1	23	150	43	978968.7	1
HT35-4	28	07.6	81	36.1	23	150	43	978968.3	1
HT37-1	28	27.2	81	43.4	23	635	43	978912.8	1
HT37-2	28	30.7	81	38.9	23	455	43	978935.2	1
HT37-3	28	21.3	81	42.4	23	310	43	978952.8	1
HT37-4	28	16.5	81	39.7	23	185	43	978964.7	1
HT38-1 NEPALGANJ DOPPLER	28	08.04	81	34.59	10	143	13	978968.0	1
HT39-1	28	01.7	82	02.0	23	220	46	978963.6	1
HT39-2	28	00.2	82	06.9	23	210	46	978963.4	1
HT39-3	27	57.7	82	13.6	23	300	46	978959.7	1
HT39-4	27	53.7	82	22.4	23	300	46	978953.0	1
HT39-5	27	53.3	82	26.5	23	300	46	978950.0	1
1983 from LAMBI E									
HT17-1	28	02.0	82	26.4	24	490	46	978876.9	1
HT17-2	28	05.8	82	21.5	24	550	46	978875.6	1

Detail Station	Position		Relia- bility Code	Elevation	Relia- bility Code	Observed g-value	No. of Ties		
	Lat N	Long E							
	°	'	°	'	metres	mGal			
HT17-3	28	06.7	82	17.6	24	610	46	978877.8	1
HT17-4	27	58.2	82	29.8	24	610	46	978880.8	1
HT17A-1	27	53.0	82	32.5	24	300	46	978940.2	1
1983 from DHANGADHI J									
HT19-1	28	47.8	80	38.9	24	200	46	978988.0	1
HT19-2	28	46.4	80	44.0	24	200	46	978988.2	1
HT19-3	28	44.9	80	49.6	24	200	46	978985.3	1
HT19-4	28	43.4	80	57.1	24	200	46	978984.6	1
HT19-5	28	38.6	81	02.9	24	200	46	978983.7	1
HT19-6	28	38.2	81	06.8	24	200	46	978983.5	1
HT19-7	28	38.3	81	16.5	24	200	46	978983.7	1
HT20-1	28	49.5	80	31.0	24	80	46	978994.3	1
HT20-2	28	50.8	80	26.1	24	80	46	978995.5	1
HT20-3	28	54.0	80	20.9	24	90	46	978996.8	1
HT20-4	28	57.0	80	15.2	24	80	46	979006.4	1
HT20-5	28	59.6	80	07.4	24	100	46	979009.5	1
HT33-1	28	52.9	80	35.0	23	300	46	978976.7	2
HT33-2	29	00.0	80	35.0	24	1220	46	978815.4	2
HT33-3	29	05.0	80	35.0	24	1726	36	978734.6	2
HT34-1	28	41.0	80	38.5	23	170	46	978989.6	1
HT34-2	28	42.2	80	34.0	23	170	46	978993.1	1
HT36-3	29	00.7	80	37.7	23	1313	36	978800.1	1
HT36-4	29	04	80	37	23	1290	46	978781.3	1
HT36-5	29	08	80	36	23	1330	46	978798.4	1
HT36-6	29	12	80	36	23	1860	46	978692.2	1
HT36-7	29	16	80	38	23	1950	46	978695.5	1
HT36-8	29	18.0	80	35.3	23	1980	46	978701.3	1
HT36-9	29	21	80	36	23	1680	46	978754.6	1
HT36-10	29	25	80	37	23	2120	46	978639.7	1
1983 from LAMOSANGU E									
HT26-1 EDR 7008	27	44.3	85	50.0	23	1327.4	32	978650.5	1
HT26-2 EDR 7014	27	43.1	85	51.5	23	1742.4	32	978574.4	1
HT26-3 EDR 7021	27	41.0	85	54.5	23	2204.8	32	978481.9	1
HT26-4	27	43.0	85	57.2	23	2440	46	978417.8	1
HT26-5 EDR 7219	27	41.5	85	58.5	23	1680	46	978471.2	1
HT26-6 EDR 7230	27	42.3	86	00.4	23	1920	46	978532.8	1
1983 from CHARIKOT C									
HT27-1 EDR 7401	27	37.5	86	04.5	23	1240.5	32	978662.8	1
HT27-2 EDR 7404	27	37.7	86	04.4	23	1064.9	32	978696.0	1
HT27-3 EDR 7508	27	37.5	86	05.6	23	1125.3	32	978685.2	1
HT27-4 EDR 7515	27	38.4	86	06.4	23	1646.3	32	978586.0	1
HT27-5 EDR 7533	27	39.0	86	09.8	23	2174.2	32	978482.3	1
HT27-6 EDR 7702	27	38.1	86	11.3	23	2476.1	32	978419.9	1
HT27-7 EDR 7732	27	37.6	86	12.9	23	2150.0	42	978480.7	1
HT27A-1	27	39.5	86	02.4	24	1830	46	978524.2	4
1983 from MUGLIN D									
HT28-1 BM 101-103	27	52.7	84	32.5	24	610	46	978862.0	1

Detail Station	Position		Relia- bility Code	Elevation	Relia- bility Code	Observed g-value	No. of Ties		
	Lat N	Long E							
	°	'	°	'	metres	mGal			
HT28-2 BM 101-111	27	55.9	84	28.9	24	370	46	978868.2	1
HT28-3	27	54.6	84	32.7	24	300	46	978868.0	1
HT28-4	27	56.4	84	33.3	24	330	46	978861.7	1
HT28-5	27	58.4	84	34.6	24	790	46	978837.8	1
HT28-6	27	58.2	84	37.0	24	850	46	978783.2	1
HT28-7	28	00.1	84	37.2	24	1040	46	978716.3	1
HT32-1 BM 101-120	27	57.7	84	24.6	23	326	10	978859.2	1
HT32-2 BM 13	27	59.4	84	25.8	23	610	46	978852.2	1
HT32-3	28	02.3	84	27.6	23	610	46	978842.4	1
HT32-5 BM 101-125	27	58.1	84	21.2	23	567	10	978818.5	1
HT32-6 BM 101-133	27	58.2	84	17.6	23	256	10	978881.5	1
HT44-1 BM 101-141	28	12.1	83	59.0	24	409	10	978855.2	1
HT44-2 BM 101-153	28	00.5	84	09.2	24	404	10	978859.7	1
HT44-3 BM 101-161	28	05.3	84	05.1	24	446	10	978849.9	1
HT44-4 BM 101-170	28	09.2	84	03.9	24	80	46	978824.1	1
HT44-5 BM 101-178	28	11.3	84	02.0	24	70	46	978803.0	1
1983 from BUTWAL E									
HT29-1	27	31.6	83	08.2	24	100	46	978918.5	1
HT29-2	27	29.2	83	11.1	24	120	46	978917.3	1
HT29-3	27	27.8	83	13.7	24	90	46	978915.6	1
HT29-4	27	44.7	83	02.7	24	140	46	978920.3	1
HT30-1	27	36.9	83	57.7	24	180	46	978892.4	1
HT30-2	27	37.1	84	00.9	24	170	46	978889.6	1
HT30-3	27	37.8	84	06.1	24	180	46	978883.3	1
HT30-4	27	40.0	84	10.8	24	180	46	978885.7	1
HT30-5	27	41.4	84	16.7	24	180	46	978888.9	1
HT30-6	27	42.6	84	22.6	24	180	46	978889.2	1
1983 from POKHARA JA									
HT31-1	28	11.8	83	53.9	24	1220	46	978742.0	1
HT31-2	28	07.9	83	57.7	24	910	46	978745.2	1
HT31-3	28	04.9	83	50.9	24	940	46	978801.3	1
HT31-4	28	01.9	83	47.3	24	760	46	978820.9	1
HT31-5	27	59.0	83	46.4	24	680	46	978828.6	1
HT31-6	27	54.0	83	38.3	24	420	46	978893.7	1
HT31-7	27	51.8	83	39.2	24	720	46	978839.2	1
HT31-8	27	56.4	83	40.5	24	580	46	978838.1	1
HT31-9	27	58.0	83	43.4	24	1010	46	978825.2	1
HT31A-1	28	14.8	83	59.5	22	910	46	978758.3	4
1983 from DHARAN F									
HT40-1 BM 104-308	26	37.0	87	10.1	23	90	10	978870.8	1
HT40-2 BM 104-295	26	35.9	87	03.6	23	82	10	978872.1	1
HT40-3 BM 104-281	26	31.3	86	56.8	23	76	10	978875.8	1
HT40-4 BM 104-264	26	36.9	86	54.3	23	88	10	978876.1	1
HT40-5 BM	26	43.7	86	56.1	23	60	46	978877.9	1
HT40-6 BM 102-253	26	38.8	86	49.5	23	117	10	978874.9	1
HT40-7 BM 104-224	26	39.4	86	35.1	23	120	10	978868.4	1
HT40-8	26	42.1	86	30.7	23	90	46	978873.5	1
HT40-9 RAJBIRAJ DOPPLER	26	32.73	86	45.06	10	82.0	10	978881.7	1
HT40-10 BM 109-036	26	30.4	86	51.4	23	70	46	978884.2	1

Detail Station	Position				Relia- bility Code	Elevation	Relia- bility Code	Observed g-value	No. of Ties
	Lat N	Long E							
	°	'	°	'		metres		mGal	
HT41-1 BM 104-440	26	39.3	87	51.0	23	130	46	978851.8	1
HT41-2 BM 104-449	26	38.6	87	56.7	23	119	10	978858.3	1
HT41-3 BM 106-001	26	37.3	87	59.2	23	114	10	978861.0	1
HT41-4 BM 106-009	26	35.0	88	03.3	23	96	10	978876.7	1
HT41-5 BM 106-015	26	33.2	88	05.6	23	86	10	978886.8	1
HT41-6 BM 104-461	26	39.2	88	03.1	23	127	10	978859.7	1
HT41-7	26	39.0	88	09.8	23	80	46	978873.9	1
HT41-8 BM 104-433	26	38.6	87	47.6	23	130	10	978847.0	1
HT42-1 BM 108-005	26	42.3	87	17.0	23	150	10	978856.5	1
HT42-2 BM 107-001	26	38.6	87	16.4	23	100	10	978862.7	1
HT42-3 BM 107-009	26	34.9	87	17.0	23	87	10	978869.8	1
HT42-4 BM2006-012	26	28.4	87	22.9	23	70	46	978873.2	1
HT42-5 BM 25-01	26	33.1	86	42.4	23	90	46	978875.5	1
HT43-1 BM 104-389	26	34.9	87	22.7	23	138	10	978850.4	1
HT43-2 BM 104-405	26	39.7	87	29.6	23	144	10	978846.8	1
HT43-3 BM 104-413	26	39.9	87	36.6	23	121	10	978849.0	1
HT43-4 BM 104-421	26	39.9	87	41.1	23	127	10	978848.6	1
HT43-5	26	51.5	87	19.4	23	1070	46	978738.2	1
HT43-6	26	53.0	87	19.3	23	910	46	978649.2	1
HT43-8	26	53.6	87	19.5	23	820	46	978710.0	1
HT43-9	26	54.2	87	18.8	23	210	46	978802.1	1
HT43-10	26	55.0	87	19.0	23	300	46	978859.6	1
1983 from HETAUDA D									
HT45-1 BM 103-237(+2.5m)	27	27	84	56	24	370	46	978844.0	1
HT45-2 BM 103-223(+0.4m)	27	31	84	50	24	300	46	978861.2	1
HT45-3 BM 103-215(+5m)	27	33	84	48	24	300	46	978865.8	1
HT45-4 BM 103-203(+1m)	27	36	84	43	24	270	46	978870.3	1
HT45-5 BM 103-194	27	36	84	38	24	210	46	978872.5	1
HT45-6 BM 103-184	27	38	84	30	24	190	46	978878.7	1
HT45-7	27	39	84	22	24	190	46	978887.1	1
HT45-8	27	36	84	16	24	170	46	978887.3	1
HT45-9	27	34	84	14	24	150	46	978888.2	1
HT47-1 BM 100-052(+0.5m)	27	19	85	00	24	490	46	978786.2	1
HT47-2 BM 100-041(+0.5m)	27	15	84	59	24	270	46	978850.2	1
HT47-3 BM 100-023	27	10	84	58	24	140	46	978862.8	1
HT47-4	27	05	84	55	24	100	46	978870.1	1
HT47-5	27	02	84	50	24	90	46	978873.6	1
HT47-6	27	02	84	57	24	100	46	978870.3	1
HT47-7	27	02	85	00	24	100	46	978867.6	1
HT47-8	27	16	84	59	24	270	46	978843.4	1
HT48-1 BM 104-005	27	12	85	01	24	150	10	978859.9	1
HT48-2 BM 104-014	27	11	85	08	24	156	10	978856.5	1
HT48-3 BM 104-024	27	10	85	14	24	147	10	978854.5	1
HT48-4 BM 104-029	27	09	85	17	24	147	10	978855.3	1
HT48-5 BM 104-035	27	07	85	22	24	128	10	978855.1	1
HT48-6	27	03	85	20	24	100	46	978857.6	1
HT48-7	26	57	85	19	24	90	46	978863.7	1
HT48-8	26	52	85	18	24	80	46	978864.8	1
HT48-9	26	47	85	17	24	70	46	978871.3	1
HT48-10	26	53	85	13	24	79.2	10	978865.9	1
HT49-1 BM 104-044	27	07	85	27	24	134	10	978858.6	1

Detail Station	Position		Relia- bility Code	Elevation	Relia- bility Code	Observed g-value	No. of Ties
	Lat N	Long E					
	°	'	°	'	metres	mGal	
HT49-2 BM 104-080	27	05	85 34	24	142	10	978853.9 1
HT49-3 BM 104-090	27	04	85 37	24	157	10	978852.9 1
HT49-4 BM2009-005	27	02	85 35	24	120	46	978857.5 1
HT49-5 BM2009-017	26	55	85 33	24	90	46	978860.9 1
HT49-6	26	51.5	85 34.2	10	81	13	978861.6 1
HT49-7 BM 104-101	27	01	85 44	24	184	10	978846.2 1
HT49-8 BM 104-109	27	00	85 50	24	208	10	978843.0 1
HT49-9 BM 104-118	26	58	85 51	24	206	10	978844.1 1
HT49-10 BM 104-153	26	56	85 58	24	159	10	978847.1 1
HT49-11 BM 104-166	26	53	86 05	24	130	46	978854.5 1
<u>1983 from BHARATPUR E</u>							
HT46-1	27	37	84 24	24	190	46	978885.3 1
HT46-2	27	35	84 30	24	180	46	978880.9 1
HT46-3	27	33	84 27	24	180	46	978886.4 1
HT46-4	27	33	84 20	24	170	46	978886.6 1
HT46-5	27	32	84 12	24	150	46	978889.4 1
HT46-6	27	32	84 15	24	150	46	978886.4 1
<u>1983 from DHALKE E</u>							
HT50-1	26	52	85 58	24	110	46	978853.5 1
HT50-2	26	48	85 57	24	90	46	978860.1 1
HT50-3 JANAKPUR TWR	26	43.56	85 55.42	10	75.9	10	978864.9 1
HT50-4	26	38	85 54	24	60	46	978869.8 1
HT50-5	26	40	85 50	24	60	46	978865.1 1
HT50-6	26	38	85 47	24	60	46	978866.7 1
HT51-1	26	53	86 11	24	110	46	978860.9 1
HT51-2	26	49	86 16	24	120	46	978859.2 1
HT51-3 BM2003-005	26	45	86 16	24	100	46	978865.2 1
HT51-4 SIRHA TWR	26	39.58	86 13.07	10	72	13	978872.8 1
<u>1983 from CHARIKOT C</u>							
HT27-8	27	38.2	86 13.8	23	1920	46	978529.8 1
HT27-9 EDR 7327	27	38.3	86 04.2	23	1503.8	32	978620.5 1
<u>1984 from BHAIRAWA J</u>							
JH06-1	27	30.3	83 21.7	22	100	46	978907.4 2
JH06-2	27	29.0	83 18.2	25	100	46	978913.4 2
JH06-3	27	32.6	83 02.0	22	100	46	978921.4 2
JH06-4	27	30.9	83 09.4	22	98	36	978918.0 2
JH06-5 BAHADURGANJ	27	33.22	82 51.07	10	104	13	978919.4 5
JH07-1	27	41.0	83 22.4	22	140	46	978911.6 3
JH07-2	27	41.3	83 16.7	22	140	46	978913.5 3
JH07-3	27	41.3	83 11.6	22	140	46	978913.0 3
JH07-4	27	40.8	83 06.5	22	120	46	978917.6 3
JH07-5	27	39.4	83 01.1	22	110	46	978920.8 3
JH07-6	27	39.4	82 54.6	24	120	46	978921.6 3
JH07-7	27	37.1	82 51.1	24	120	46	978918.4 3
JH07-8	27	39.1	83 28.1	22	140	46	978903.9 3
JH07-9	27	34.2	83 27.8	22	120	46	978901.0 3

Detail Station	Position					Elevation	Relia- bility Code	Observed g-value	No. of Ties
	Lat N		Long E		Relia- bility Code				
	•		•			metres		mGal	
1984 from BUTWAL D									
JH09-1	27	45.0	83	30.5	22	150	45	978888.5	2
JH09-2	27	47.8	83	31.7	22	610	45	978843.1*	2
JH09-3	27	51.0	83	33.7	22	1020	45	978818.1	2

GRAVITY STATION POSITION AND ELEVATION ACCURACY CODES

In the gravity station data lists presented with this report the position and elevation value quoted for each gravity station is followed by a two-digit numerical code. This code indicates the source and accuracy of the quoted position or elevation value in accordance with the following tables.

PLAN POSITION CODES

Source Type	Code	Source Class
Survey control	10	Ground survey of third order accuracy or better.
	11	Inertial or Satellite positioning systems of survey accuracy.
	12	Positioning systems of navigational accuracy.
Map control	20	Map scale larger than 1:24,000
	21	1:24,000 or 1:25,000
	22	1:50,000 or 1:63,360
	23	1:100,000 or 1:126,720
	24	1:250,000 or 1:253,440
	25	smaller than 1:253,440
Air Photo control	30	Photo scale larger than 1:10,000
	31	1:10,000 to 1:60,000
	32	smaller than 1:60,000
Astro control	40	Astro position of survey accuracy.

ELEVATION CODES

Source Type	Code	Source Class			
Survey control	10	Bench mark in level nets of third order accuracy or better.			
	11	Bench mark in local or engineer level nets.			
	12	Inertial or Satellite positioning systems of survey accuracy.			
	13	Vertical Angle heighting of, or related to, a survey control net.			
	14	Barometric heighting from survey control of a higher accuracy.			
Map control	20				
Checked Spot	to	Code	Contour Interval of Map		
Elevation	26	20,30,40	less than 3 metres		(10 feet)
Map control	30	21,31,41	from 3 to 6 metres	(10 to 19 feet)	
Unchecked Spot	to	22,32,42	from 7 to 12 metres	(20 to 39 feet)	
Elevation	36	23,33,43	from 13 to 18 metres	(40 to 59 feet)	
Map control	40	24,34,44	from 19 to 24 metres	(60 to 79 feet)	
Contour	to	25,35,45	from 25 to 30 metres	(80 to 100 feet)	
Interpolation	46	26,36,46	greater than 30 metres	(100 feet)	

GRAVITY STATION POSITION AND ELEVATION ACCURACY CODES

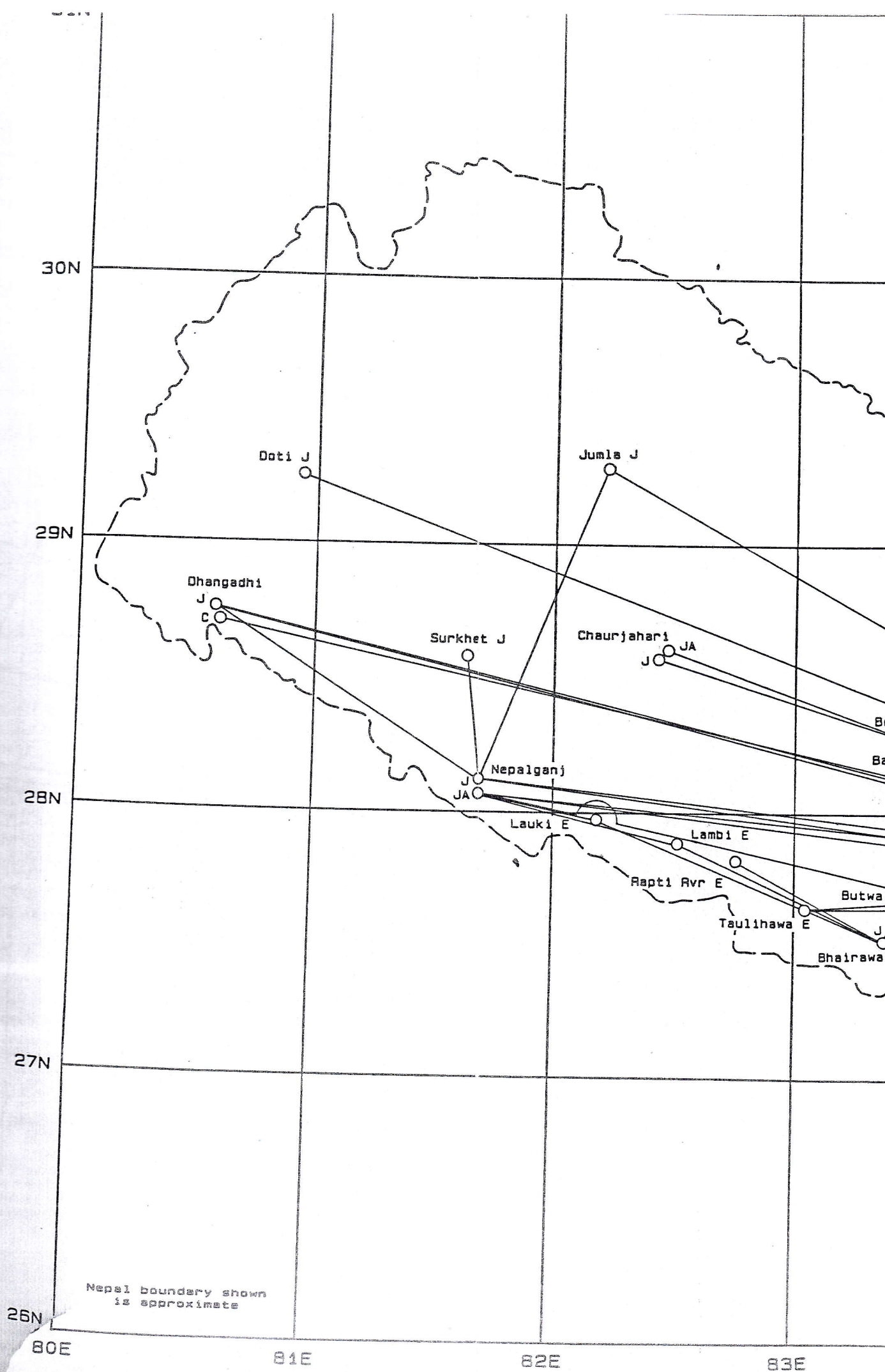
In the gravity station data lists presented with this report the position and elevation value quoted for each gravity station is followed by a two-digit numerical code. This code indicates the source and accuracy of the quoted position or elevation value in accordance with the following tables.

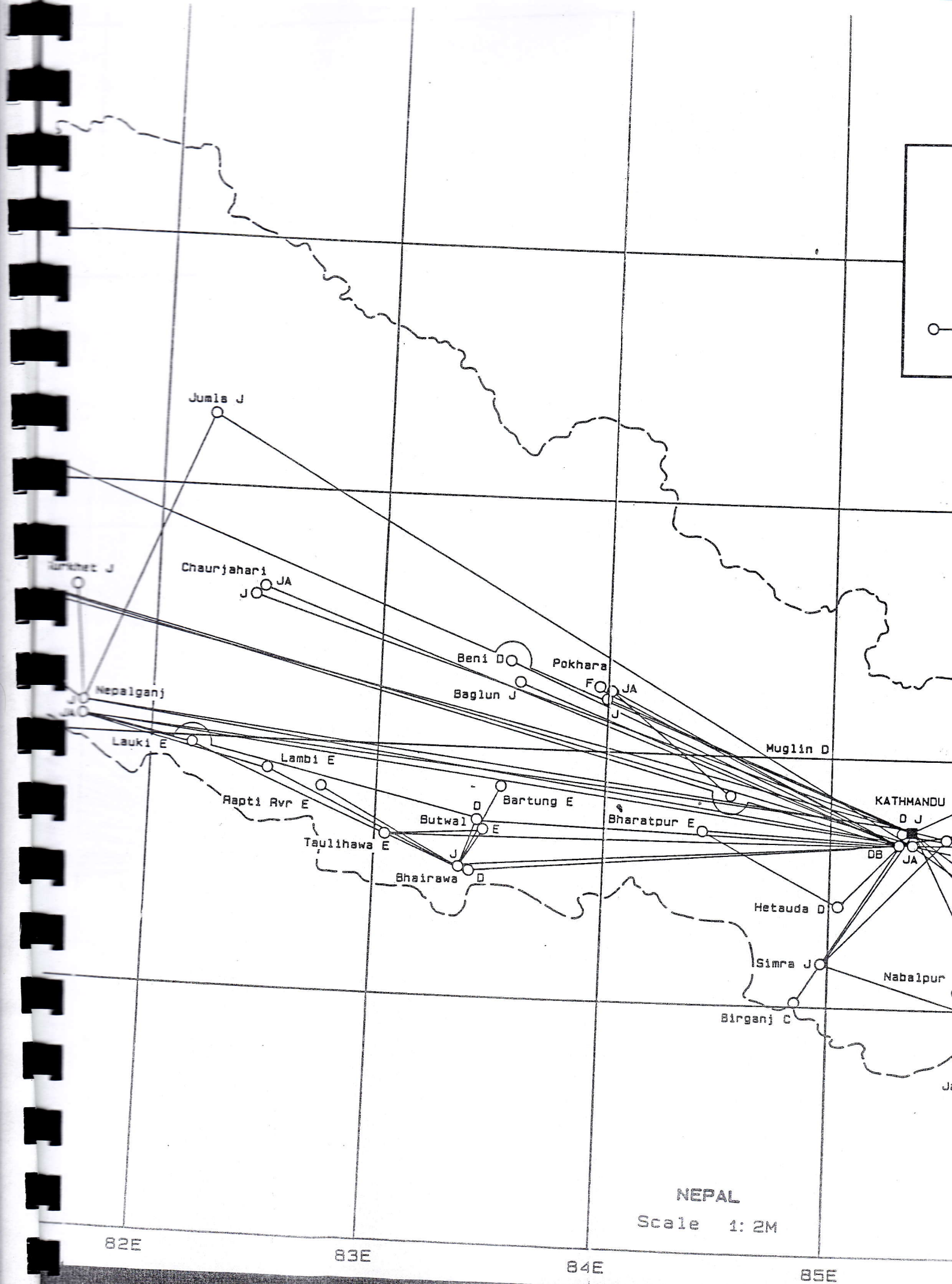
PLAN POSITION CODES

Source Type	Code	Source Class
Survey control	10	Ground survey of third order accuracy or better.
	11	Inertial or Satellite positioning systems of survey accuracy.
	12	Positioning systems of navigational accuracy.
Map control	20	Map scale larger than 1:24,000
	21	1:24,000 or 1:25,000
	22	1:50,000 or 1:63,360
	23	1:100,000 or 1:126,720
	24	1:250,000 or 1:253,440
	25	smaller than 1:253,440
Air Photo control	30	Photo scale larger than 1:10,000
	31	1:10,000 to 1:60,000
	32	smaller than 1:60,000
Astro control	40	Astro position of survey accuracy.

ELEVATION CODES

Source Type	Code	Source Class
Survey control	10	Bench mark in level nets of third order accuracy or better.
	11	Bench mark in local or engineer level nets.
	12	Inertial or Satellite positioning systems of survey accuracy.
	13	Vertical Angle heighting of, or related to, a survey control net.
	14	Barometric heighting from survey control of a higher accuracy.
Map control Checked Spot Elevation	20 to 26	Code Contour Interval of Map 20,30,40 less than 3 metres (10 feet)
Map control Unchecked Spot Elevation	30	21,31,41 from 3 to 6 metres (10 to 19 feet)
	to	22,32,42 from 7 to 12 metres (20 to 39 feet)
	36	23,33,43 from 13 to 18 metres (40 to 59 feet)
Map control Contour Interpolation	40	24,34,44 from 19 to 24 metres (60 to 79 feet)
	to	25,35,45 from 25 to 30 metres (80 to 100 feet)
	46	26,36,46 greater than 30 metres (100 feet)



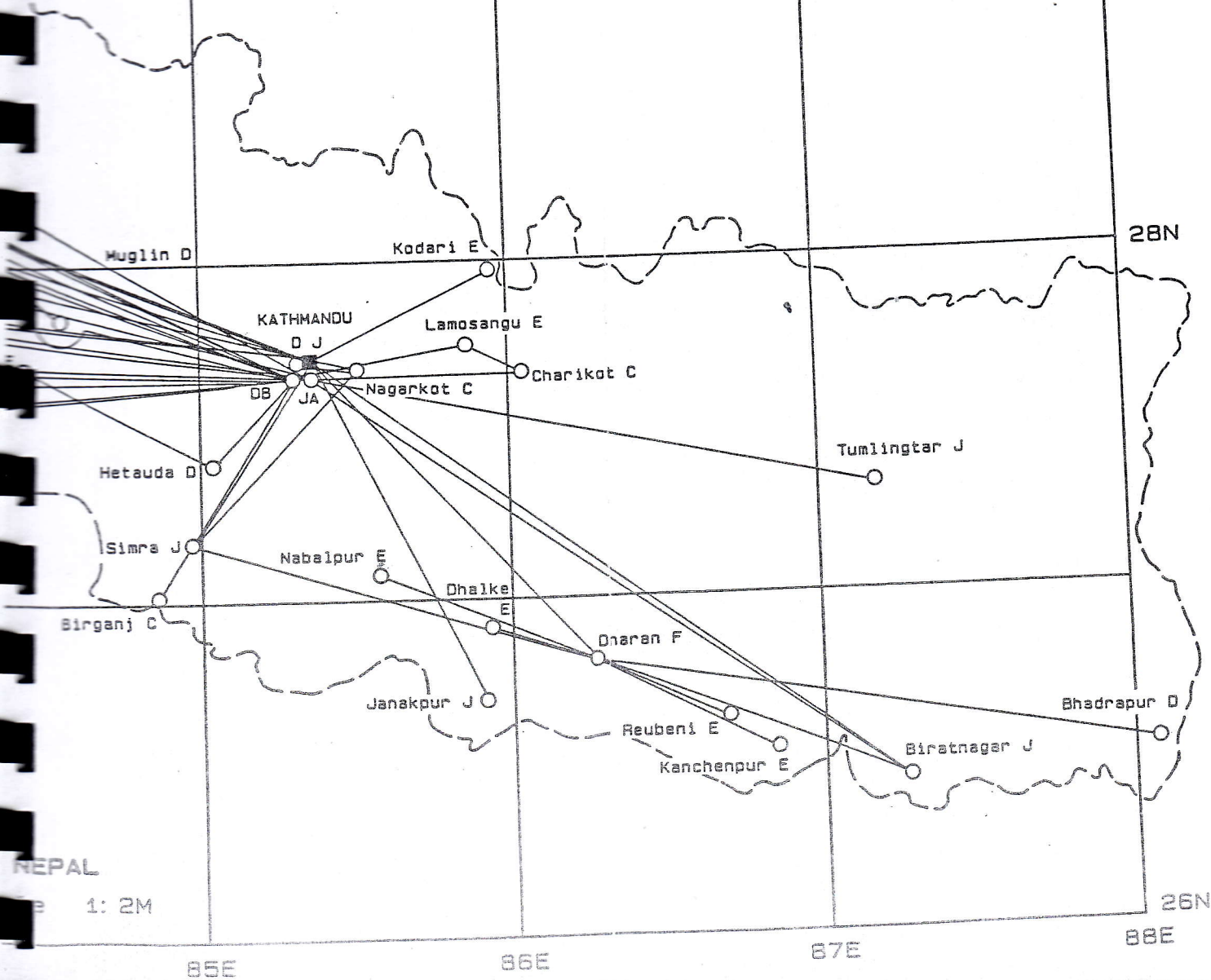


NEPAL GRAVITY SURVEYS 1981-84

Diagram of Gravity Base Station Net

Fundamental station KATHMANDU J

○—○ Independent connection observed with 2 or more LR gravimeters.



REPORT AND RESULTS OF
GRAVITY SURVEYS IN NEPAL
1981-84

DIRECTORATE OF MILITARY SURVEY
MINISTRY OF DEFENCE UK

TWB
Geodesy Division
Mapping and Charting Establishment RE
December 1985

Reference: MCE/Caj

FOREWORD

This report, together with its companion report on the geodetic survey, gives in considerable detail the results of geodetic and gravity surveys carried out in the Kingdom of Nepal between 1981 and 1984 by military surveyors from 512 Specialist Team Royal Engineers and 19 Topographic Squadron Royal Engineers. The work was done in close cooperation with the Nepal Survey Department, and has provided a lasting high accuracy framework through the Kingdom on which any future survey and mapping work can be based.

The geodetic survey may well be the last national framework to be observed in the "classical" manner with theodolites and distance measuring equipment, for satellite based and other techniques will soon supersede earth based geometric figures in quality of results and obtainable at greatly reduced costs and time.

The mathematical adjustments presented in these reports are the work of the civilian staff of the Mapping and Charting Establishment Royal Engineers. The reports and analysis have been prepared by Mr T W Bassett, under whose expert guidance the work has been painstakingly carried out.

The high quality of the results of the surveys are a tribute to the demanding standards which were set and maintained both in the fieldwork and its subsequent adjustment, and the close cooperation between the surveyors of the United Kingdom and the Kingdom of Nepal.

C N THOMPSON
Major General
Director of Military Survey
United Kingdom

July 1986

SUMMARY

This report presents the results of gravity surveys made in Nepal by British military survey units during the period 1981-84

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A	Data List of Gravity Base Stations
B	Data List of Gravity Detail Stations
C	Gravity Station Position and Elevation Accuracy Codes
D	Diagram of Gravity Base Station Net
E	Diagram of Distribution of Gravity Detail Stations
F	Description Sheets for Gravity Base Stations

NEPAL GRAVITY SURVEYS 1981-84

DESCRIPTIONS OF GRAVITY BASE STATIONS

BAGLUN J	KATHMANDU C
BARTUNG E	KATHMANDU CA
BENI D	KATHMANDU CB
BHADRAPUR D	KATHMANDU CC
BHAIRAWA D	KATHMANDU D
BHAIRAWA J	KATHMANDU DA
BHARATPUR E	KATHMANDU DB
BIRGANJ C	KATHMANDU F
BIRATNAGAR J	KATHMANDU J
BUTWAL D	KATHMANDU JA
BUTWAL E	KATHMANDU K
CHARIKOT C	LAMBI E
CHAURJAHARI J	LAMOSANGU E
CHAURJAHARI JA	LAUKI E
DHALKE E	MUGLIN D
DHANGADHI C	NABALPUR E
DHANGADHI J	NAGARKOT C
DHARAN F	NEPALGANJ J
DOTI J	NEPALGANJ JA
HETAUDA D	POKHARA F
JANAKPUR J	POKHARA J
JUMLA J	POKHARA JA
KANCHANPUR E	RAPTI RIVER E
KODARI E	REUBENI E
	SIMRA J
	SURKHET J
	TAULIHAWA E
	TUMLINGTAR J

RUPANI